

Assignment No:3

Memory Management Methods

Q.1 Write the simulation program for demand paging and show the page scheduling and total number of page faults according the optimal page replacement algorithm. Assume the memory of 'n' frames.

Reference String: 8, 5, 7, 8, 5, 7, 2, 3, 7, 3, 5, 9, 4, 6, 2.

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int ref[] = {8,5,7,8,5,7,2,3,7,3,5,9,4,6,2};
```

```
    int n = 15, frames, f[10], faults = 0;
```

```
    int i, j, k, pos, far, found;
```

```
    printf("Enter number of frames: ");
```

```
    scanf("%d", &frames);
```

```
    for(i = 0; i < frames; i++)
```

```
        f[i] = -1;
```

```
    printf("\nPage\tFrames\n");
```

```
    for(i = 0; i < n; i++) {
```

```
        found = 0;
```

```
        for(j = 0; j < frames; j++)
```

```
            if(f[j] == ref[i])
```

```
                found = 1;
```

```
        if(!found) {
```

```
            faults++;
```

```
            for(j = 0; j < frames; j++)
```

```
                if(f[j] == -1) {
```

```
                    pos = j;
```

```
                    break;
```

```
                }
```

```
            if(j == frames) {
```

```
                far = -1;
```

```
                pos = 0;
```

```

        for(j = 0; j < frames; j++) {
            for(k = i + 1; k < n && f[j] != ref[k]; k++);
            if(k > far) {
                far = k;
                pos = j;
            }
        }
        f[pos] = ref[i];
    }
    printf("%d\t", ref[i]);
    for(j = 0; j < frames; j++)
        printf("%d ", f[j]);
    printf("\n");
}
printf("\nTotal Page Faults = %d\n", faults);
return 0;
}

```

Q.2 Write the simulation program to implement demand paging and show the page scheduling and total number of page faults according to the LRU (using counter method) page replacement algorithm. Assume the memory of n frames.

Reference String : 3,5,7,2,5,1,2,3,1,3,5,3,1,6,2

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int ref[] = {3,5,7,2,5,1,2,3,1,3,5,3,1,6,2};
```

```
    int n = 15, frames, f[10], count[10];
```

```
    int i, j, pos, faults = 0, found, min;
```

```
    printf("Enter number of frames: ");
```

```

scanf("%d", &frames);
for(i = 0; i < frames; i++) {
    f[i] = -1;
    count[i] = 0;
}
printf("\nPage\tFrames\n");
for(i = 0; i < n; i++) {
    found = 0;
    for(j = 0; j < frames; j++) {
        if(f[j] == ref[i]) {
            found = 1;
            count[j] = i;    // Counter method
            break;
        }
    }
    if(!found) {
        faults++;
        pos = -1;
        for(j = 0; j < frames; j++)
            if(f[j] == -1) {
                pos = j;
                break;
            }
        if(pos == -1) {
            min = count[0];
            pos = 0;
            for(j = 1; j < frames; j++)
                if(count[j] < min) {
                    min = count[j];
                    pos = j;
                }
        }
    }
}

```

```

        f[pos] = ref[i];
        count[pos] = i;
    }
    printf("%d\t", ref[i]);
    for(j = 0; j < frames; j++)
        printf("%d ", f[j]);
    printf("\n");
}
printf("\nTotal Page Faults = %d\n", faults);
return 0;
}

```

Q.3 Write the simulation program for demand paging and show the page scheduling and total number of page faults according the MFU page replacement algorithm. Assume the memory of n frames.

Reference String: 8, 5, 7, 8, 5, 7, 2, 3, 7, 3, 5, 9, 4, 6, 2

```

#include <stdio.h>

int main()
{
    int ref[] = {3,5,7,2,5,1,2,3,1,3,5,3,1,6,2};
    int n = 15, frames, f[10], count[10];
    int i, j, pos, faults = 0, found, min;
    printf("Enter number of frames: ");
    scanf("%d", &frames);
    for(i = 0; i < frames; i++) {
        f[i] = -1;
        count[i] = 0;
    }
    printf("\nPage\tFrames\n");
    for(i = 0; i < n; i++) {
        found = 0;
        for(j = 0; j < frames; j++) {

```

```

        if(f[j] == ref[i]) {
            found = 1;
            count[j] = i;    // Counter method
            break;
        }
    }
    if(!found) {
        faults++;
        pos = -1;
        for(j = 0; j < frames; j++)
            if(f[j] == -1) {
                pos = j;
                break;
            }
        if(pos == -1) {
            min = count[0];
            pos = 0;
            for(j = 1; j < frames; j++)
                if(count[j] < min) {
                    min = count[j];
                    pos = j;
                }
        }
        f[pos] = ref[i];
        count[pos] = i;
    }
    printf("%d\t", ref[i]);
    for(j = 0; j < frames; j++)
        printf("%d ", f[j]);
    printf("\n");
}
printf("\nTotal Page Faults = %d\n", faults);

```

```

    return 0;
}

```

Q.4 Write the simulation program for demand paging and show the page scheduling

and total number of page faults according the FIFO page replacement algorithm.

Assume the memory of n frames.

Reference String: 3, 4, 5, 6, 3, 4, 7, 3, 4, 5, 6, 7, 2, 4, 6 .

```

#include <stdio.h>

int main()
{
    int ref[] = {3,4,5,6,3,4,7,3,4,5,6,7,2,4,6};
    int n = 15, frames, f[10];
    int i, j, pos = 0, faults = 0, found;
    printf("Enter number of frames: ");
    scanf("%d", &frames);
    for(i = 0; i < frames; i++)
        f[i] = -1;
    printf("\nPage\tFrames\n");
    for(i = 0; i < n; i++) {
        found = 0;
        for(j = 0; j < frames; j++)
            if(f[j] == ref[i])
                found = 1;
        if(!found) {
            f[pos] = ref[i];
            pos = (pos + 1) % frames;
            faults++;
        }
        printf("%d\t", ref[i]);
        for(j = 0; j < frames; j++)
            printf("%d ", f[j]);
    }
}

```

```
    printf("\n");  
}  
printf("\nTotal Page Faults = %d\n", faults);  
return 0;  
}
```